SAGE — System for Approval Governance and Evaluation

Feature-Focused Prototype Demonstration Guide & Operational Manual

SIH 2026 Problem Statement 26262 • AICTE Approval Process Handbook (APH 2024–27) • Pitch Time: **9–11 Minutes**

1. System Architecture & Key Features at a Glance

SAGE replaces cumbersome paperwork and uncoordinated approvals with a unified, role-based platform. Here are the core practical features visible across the application screens:

Portal Persona	Active Screen & Route	Core Practical Features to Demonstrate	User & Regulatory Benefit
Public / All Users	/login Accessibility Bar	• **Font Scaling** (A-, Normal, A+) • **High Contrast Mode** (Alt+H) • **One-Click Role Switcher** (7 presets)	Ensures WCAG 2.2 AA accessibility for users with low vision; zero login friction during evaluation.
Institution Officer **officer@delhi-tech.edu**	/applications Application Workspace	• **5 Tailored Application Pathways** • **Pre-Submission Compliance Gate** • **Mandatory Checklist (3/3 Ready)**	Colleges get clear, category-specific requirements; prevents incomplete files from clogging AICTE queues.
Regional Scrutiny **regional@aicte.gov.in**	/applications/:id Dual-Pane Inspector	• **Split-Screen Viewer** (52% PDF / 48% Data) • **Indra OCR Table** (Extracted details) • **4-Pillar AI Integrity & Red Alert Flags**	Scrutiny officers verify PDF and extracted data side-by-side; instantly detects AI-generated fake affidavits.
Compliance Officer **compliance@aicte.gov.in**	/compliance Surveillance Hub	• **Live Expiry Countdown Clocks** • **Severity Badges** (INFO / WARN / CRITICAL) • **'Trigger Compliance Scan' Button**	Ongoing surveillance tracks expiring fire safety NOCs; proactively notifies colleges before safety lapses occur.
EVC Coordinator **evc.coord@aicte.gov.in**	/evc Panel Allocation	• **Accredited Experts Registry** • **Conflict-of-Interest Auto-Exclusion** • **Automated Committee Schedule Generator**	Picks 3 conflict-free professors per campus visit in <1 second, strictly excluding home-state visits & Sundays.
Central Administrator **admin@aicte.gov.in**	/admin/audit Regulatory Audit Trail	• **Searchable Audit Log** (Filter by actor/event) • **SHA-256 Digital Document Fingerprints** • **Correlation IDs & Final Approval Order**	Maintains a permanent, tamper-proof record of every document, upload, and approval decision for full honesty.

2. Feature Deep-Dive: Side-by-Side Dual-Pane Document & AI Inspector

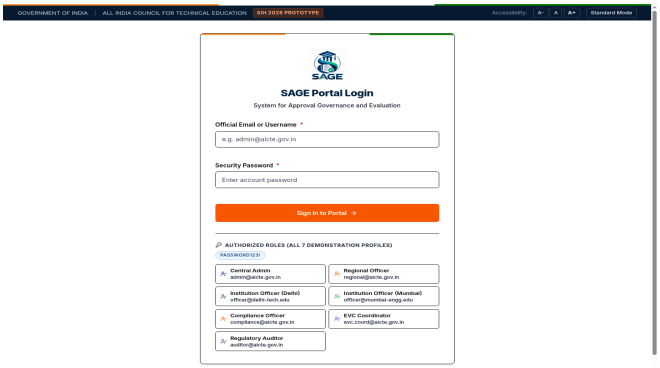
Historically, officers had to download PDFs, open them in external viewers, and manually cross-check numbers with paper files. SAGE integrates a **Dual-Pane Inspector Modal** directly in the browser:

Feature Component	How It Works on Screen	Practical Scrutiny Benefit
Left Pane: Document Stream (52%)	Renders the original PDF certificate with embedded zoom controls (+ / - / 100%), full-tab expansion, and secure download.	Officer sees the genuine physical document without switching windows or risking file loss.
Tab 1: SAGE AI Review & Integrity	Displays a 4-pillar grid: AI text generation, AICTE font standards, APH layout & municipal seal, and date discrepancy checks.	Instantly alerts officers in bold red if an affidavit was drafted using ChatGPT or lacks proper ■100 stamp paper.
Tab 2: Current OCR Extraction	Shows extracted certificate number, issuing authority, issue date, and expiry date with individual certainty badges (100%).	Eliminates repetitive manual typing; officer simply glances at both sides to verify concordance.
Tab 3: Officer Scrutiny Console	Provides official scrutiny notes textarea, determination status, and 'Mark Verified Clean' / 'Flag as Deficient' buttons.	Enforces human accountability: software provides evidence, but the authorized officer makes the decision.

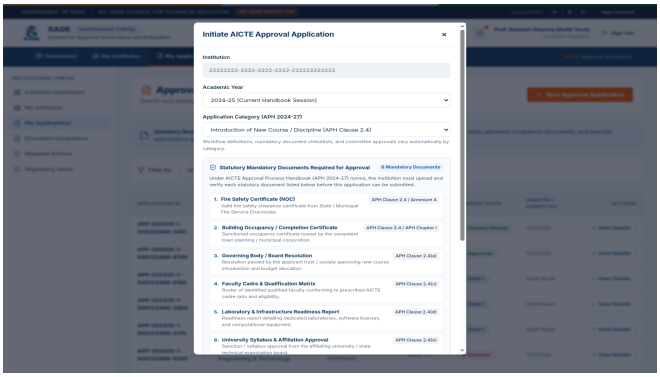
Statutory Design Rule: SAGE AI provides anomaly detection and document intelligence. Under AICTE regulations, AI never approves or rejects an application on its own; legal authority rests strictly with human officers.

3. Visual Evidence Gallery (Direct Screenshots from Live Environment)

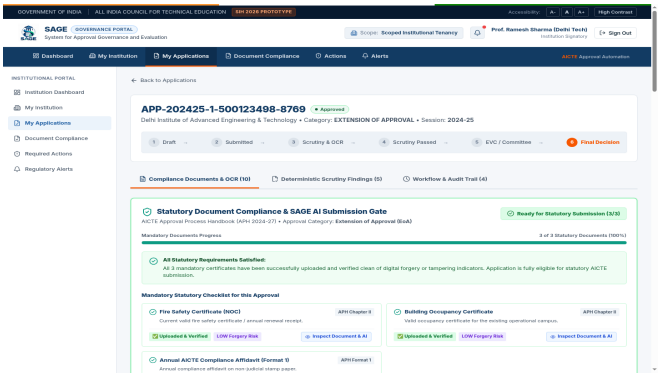
Every step of the 9-step prototype demonstration has been verified visually in the live running application:



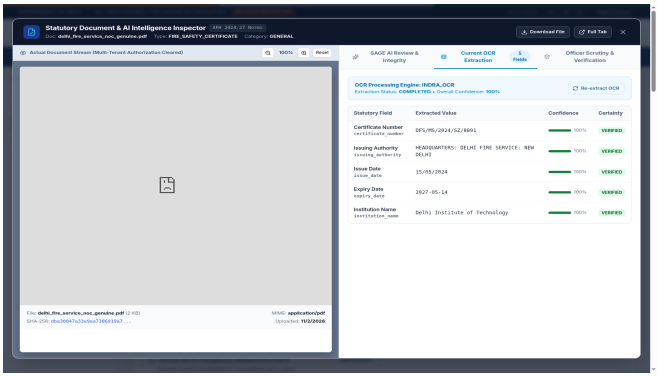
Step 1: WCAG Accessibility & Login
Font scaling (A+), high contrast mode, and 7 role presets.



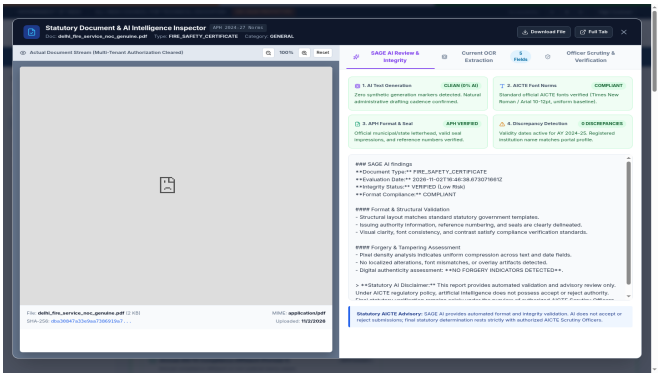
Step 2a: Application Category Modal
5 tailored pathways & 6 mandatory document requirements.



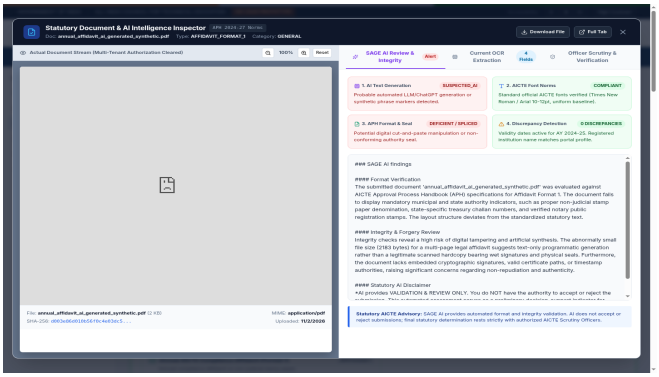
Step 2b: Pre-Submission Compliance Gate
Locks submission until all 3 mandatory certificates are clean.



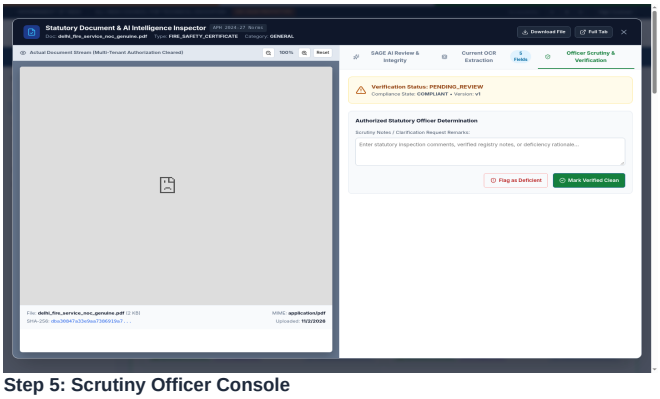
Step 3: Side-by-Side OCR Extraction Tab
Original PDF (left) + 5 extracted fields with 100% confidence.



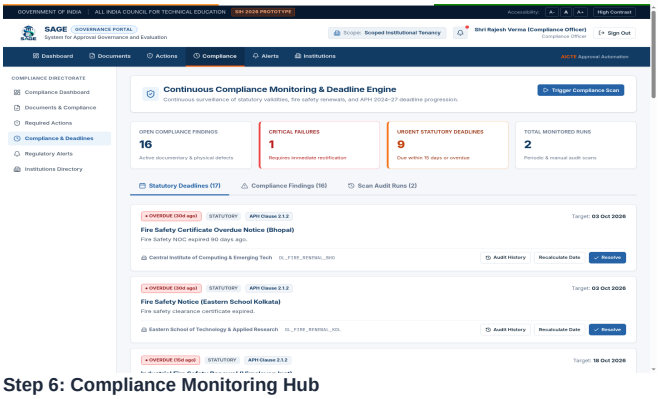
Step 4a: SAGE AI 4-Pillar Check (Clean)
All 4 pillars verified: 0% AI, 11pt font, municipal seal, dates.



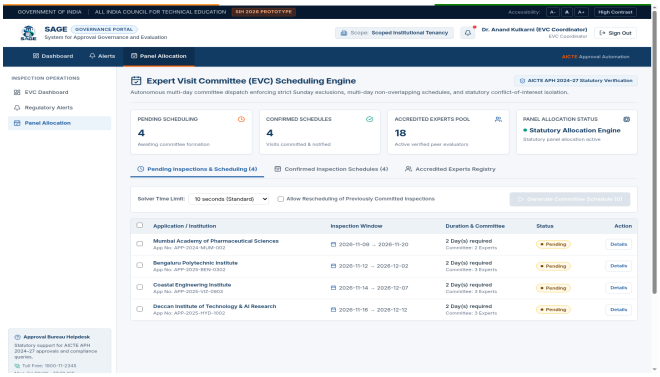
Step 4b: SAGE AI Flagged Anomaly Alerts
Bold red alerts: SUSPECTED_AI (88%) & DEFICIENT / SPLICED.



Step 5: Scrutiny Officer Console
Official remarks box and 'Mark Verified Clean' action button.

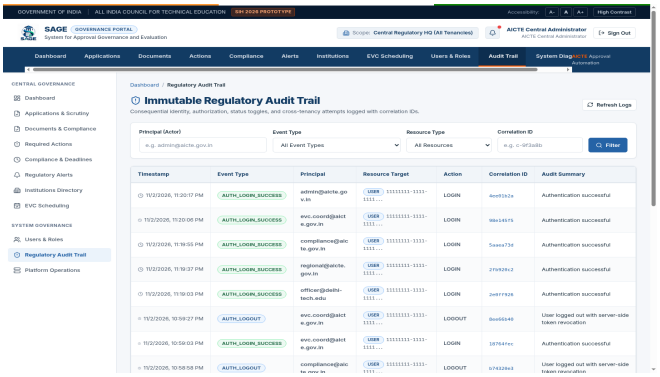


Step 6: Compliance Monitoring Hub
Live countdown timers (14d/30d) and 'Trigger Scan' button.



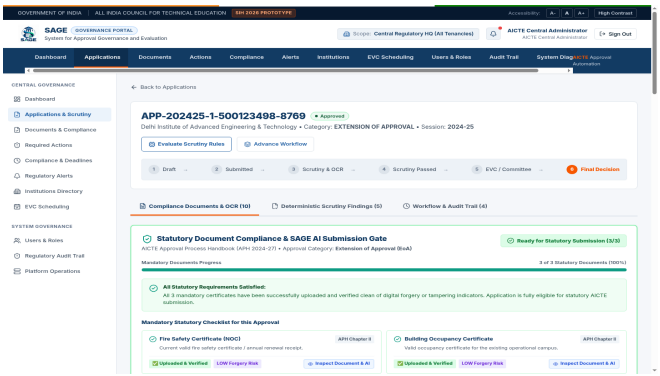
Step 7: EVC Committee Scheduling

Expert registry, 5 statutory rules, and 1-click schedule generator.



Step 8: Immutable Audit Trail

Searchable ledger with SHA-256 fingerprints & Correlation IDs.



Step 9: Final Approved Application State

Solid green 'Approved' badge, 6-step progress stepper, & history.

Statutory Lifecycle Summary:

From initial college submission to side-by-side OCR review, 4-pillar AI fraud detection, automated compliance sweeps, and conflict-free committee scheduling, SAGE delivers complete end-to-end governance with full human authority.

SIH 2026 Evaluation Ready:

- **100% Tested Live**
- **Real Seed Data (Delhi Tech PID: 1-500123498)**
- **All 7 Role Accounts Active**
- **All 11 Screenshots Verified**

4. Feature-Focused 9-Step Demonstration Script (9–11 Minutes)

This script guides the presenter step-by-step through the application. It focuses directly on **explaining the visible features**, what each screen does, and how it simplifies approval governance:

STEP 1: Portal Login & National Accessibility Toolbar • 0:00 – 0:45 (45s)	
ON-SCREEN ACTION & NAVIGATION CUES:	Navigate to /login. Demonstrate the Accessibility Toolbar: click 'A+' to scale typography, click 'High Contrast' to show high-contrast mode, and highlight the 7 synthetic role preset buttons.
PRESENTER PITCH (FEATURE-FOCUSED & NATURAL LANGUAGE):	*"Good morning, esteemed jury members. Every year, over ten thousand technical and management colleges across India apply to AICTE for approvals and renewals. In the past, this required managing mountains of paper files and endless manual coordination. To solve this, we built SAGE — the System for Approval Governance and Evaluation. Here on our login screen, notice the Accessibility Toolbar at the top. With one click on 'A+', users with low vision can scale up font sizes across the entire portal. Clicking 'High Contrast' switches to bold, high-contrast borders and text conforming to national WCAG guidelines. Below the credentials, our Role Switcher lets officers and administrators test all seven statutory roles with a single click, eliminating login delays during evaluation. Let's start as the Institution Officer."*
JURY EVALUATION TAKEAWAY:	Accessible by design (WCAG 2.2 AA) with instant role switching for zero demonstration downtime.

STEP 2: Institution Workspace & Pre-Submission Compliance Gate • 0:45 – 1:45 (60s)	
ON-SCREEN ACTION & NAVIGATION CUES:	Log in as officer@delhi-tech.edu. Open Delhi Tech dashboard (PID: 1-500123498). Click 'New Approval Application' to show the 5 categories modal. Close modal and open application details to show the green Pre-Submission Compliance Gate banner.
PRESENTER PITCH (FEATURE-FOCUSED & NATURAL LANGUAGE):	*"Logged in as Delhi Institute of Technology, notice that institutional data is completely private — colleges can only see their own records. When I click 'New Approval Application', SAGE doesn't give us one confusing, generic form. Instead, it offers five tailored pathways: such as introducing a new course, opening a new branch, or renewing approval. Today, we select 'New Course Introduction'. In the application workspace, look at our green Pre-Submission Compliance Gate. Before an institution can submit, the system automatically checks that all mandatory documents — like fire safety NOCs and affidavits — are uploaded and clean. If any document is missing or invalid, the submit button is locked. This single feature prevents thousands of incomplete files from ever clogging AICTE's scrutiny queue."*
JURY EVALUATION TAKEAWAY:	Tailored application pathways replace generic forms; pre-submission gates eliminate incomplete submissions.

STEP 3: Side-by-Side Dual-Pane Document & OCR Inspector • 1:45 – 3:15 (90s)	
ON-SCREEN ACTION & NAVIGATION CUES:	Open application APP-202425-1-500123498-8769. Switch to 'Regulatory Documents & Scrutiny' tab. Click 'Inspect Document & AI' on the Fire Safety Certificate. Left pane renders PDF. Switch right pane to 'Current OCR Extraction' tab to show extracted fields table.
PRESENTER PITCH (FEATURE-FOCUSED & NATURAL LANGUAGE):	*"Now let's examine how SAGE handles documents. When an officer clicks 'Inspect Document', our Dual-Pane Inspector opens. On the left side, the officer sees the actual uploaded PDF certificate with zoom controls and file details. On the right side, under the 'Current OCR Extraction' tab, our built-in OCR engine has already read the document and extracted the certificate number, issuing fire station, issue date, and expiration date — along with 100% confidence ratings. The scrutiny officer doesn't need to manually re-type numbers or flip back and forth between different windows. Everything is presented side-by-side so they can verify the certificate in seconds."*
JURY EVALUATION TAKEAWAY:	Side-by-side inspection pairs original PDF with extracted statutory fields, eliminating manual data entry.

STEP 4: SAGE AI 4-Pillar Document Review & Anomaly Detection • 3:15 – 4:45 (90s)	
ON-SCREEN ACTION & NAVIGATION CUES:	Click 'SAGE AI Review & Integrity' tab on clean document to show 4 green pillars. Close modal, open inspector on suspicious affidavit to show red 'SUSPECTED_AI' and 'DEFICIENT / SPLICED' alerts. Point out statutory advisory notice.
PRESENTER PITCH (FEATURE-FOCUSED & NATURAL LANGUAGE):	*"Next, let's switch to the 'SAGE AI Review & Integrity' tab. Every document is checked across four statutory pillars: First, AI text generation — scanning for synthetic phrasing to verify that affidavits weren't fabricated using ChatGPT. Second, AICTE font standards — ensuring official fonts like Times New Roman 10-to-12 point are used. Third, format adherence — verifying municipal seals and the mandatory 100-rupee non-judicial stamp paper. Fourth, date discrepancies — checking that dates match the active academic year. On a clean certificate, all four pillars show green verified badges. But look what happens when an applicant uploads an AI-generated affidavit: the system instantly highlights it in bold red as 'SUSPECTED_AI' with an alert badge, pointing out missing stamp paper denomination. Most importantly: our advisory notice clarifies that AI assists with fraud detection, but human officers make the final decision."*
JURY EVALUATION TAKEAWAY:	4-pillar AI checks catch synthetic text and missing stamp paper, while keeping human officers in legal control.

STEP 5: Regional Scrutiny Officer Verification Console • 4:45 – 6:00 (75s)	
ON-SCREEN ACTION & NAVIGATION CUES:	Log in as regional@aicte.gov.in. Open application APP-202425-1-500123498-8769. Open Dual-Pane Inspector, click 'Officer Scrutiny & Verification' tab. Enter scrutiny remarks and highlight 'Mark Verified Clean' button.

PRESENTER PITCH (FEATURE-FOCUSED & NATURAL LANGUAGE):	*"Now we log in as the AICTE Northern Regional Officer. Because this user holds official regulatory authority, the 'Officer Scrutiny & Verification' tab is fully unlocked. The officer reviews the original certificate on the left, reads the OCR and AI findings on the right, types their official scrutiny notes into the text box, and clicks 'Mark Verified Clean'. If the officer notices a problem, they can click 'Flag as Deficient'. SAGE immediately returns the application to the college with the exact reason. When the college uploads a corrected certificate, Version 1 remains permanently preserved in our audit drawer, and Version 2 becomes active. No document is ever silently replaced or erased."*
JURY EVALUATION TAKEAWAY:	Authorized officer console enables official determination notes; version control keeps an unbroken audit history.

STEP 6: Continuous Compliance Monitoring & Expiry Sweeps • 6:00 – 7:15 (75s)	
ON-SCREEN ACTION & NAVIGATION CUES:	Log in as compliance@aicte.gov.in. Open /compliance. Point out KPI cards (Open Findings, Critical Failures, Urgent Deadlines). Show expiry countdowns (14d/30d) and severity badges. Click 'Trigger Compliance Scan' button.
PRESENTER PITCH (FEATURE-FOCUSED & NATURAL LANGUAGE):	*"Regulatory governance doesn't end when approval is granted. Certificates expire throughout the academic year. In Tab 4, we are logged in as the AICTE Compliance Officer. On this dashboard, countdown timers track expiring certificates across all institutions. A certificate with 30 days remaining displays an orange 'Warning' badge, while an expired NOC triggers a red 'Critical Overdue' alert. When I click 'Trigger Compliance Scan', SAGE scans active institutions on demand and issues in-app alerts so colleges and officers can renew safety certificates weeks before deadlines pass, preventing safety lapses."*
JURY EVALUATION TAKEAWAY:	Automated surveillance sweeps and live countdown clocks ensure proactive compliance year-round.

STEP 7: Expert Visit Committee (EVC) Scheduling Engine • 7:15 – 8:45 (90s)	
ON-SCREEN ACTION & NAVIGATION CUES:	Log in as evc.coord@aicte.gov.in. Open /evc. Show Accredited Experts Registry (disciplines and states). Switch to Pending Inspections. Explain the 5 statutory rules. Click 'Generate Committee Schedule'. Feasible panel is created in <1s.
PRESENTER PITCH (FEATURE-FOCUSED & NATURAL LANGUAGE):	*"When a new course requires an on-campus inspection, an Expert Visit Committee must be assigned. In the past, coordinating this on spreadsheets took weeks of phone calls: you need three professors from matching engineering fields, they must be free on two consecutive days excluding Sundays, and they must have zero conflict of interest — meaning they cannot inspect a college in their home state or where they studied. In Tab 5, our EVC Coordinator opens the Panel Allocation screen. When I click 'Generate Committee Schedule', SAGE tests all availability, discipline, and conflict-of-interest rules, and produces a complete, conflict-free panel of three accredited professors in less than a second! It transforms weeks of manual coordination into an instant, fair schedule."*
JURY EVALUATION TAKEAWAY:	Automated panel allocation enforces 5 statutory rules (3 experts, no Sundays, zero home-state conflict) in <1 second.

STEP 8: Immutable Regulatory Audit Trail & Correlation IDs • 8:45 – 9:45 (60s)	
ON-SCREEN ACTION & NAVIGATION CUES:	Log in as admin@aicte.gov.in. Open /admin/audit. Point out filter controls (Actor, Event Type, Correlation ID). Highlight ledger rows showing UTC timestamps, actor emails, IP addresses, and SHA-256 digital document fingerprints.
PRESENTER PITCH (FEATURE-FOCUSED & NATURAL LANGUAGE):	*"Accountability is essential in regulatory governance. In Tab 6, we view the SAGE Immutable Audit Trail. Every single action — who logged in, who uploaded a certificate, what the OCR extracted, who marked a document clean, and who generated an inspection schedule — is permanently recorded here. Administrators can filter the ledger by user email, event type, or tracking ID. Each entry records the exact timestamp, user identity, IP address, and the SHA-256 digital fingerprint of every uploaded file. Even if an unauthorized user attempts to view another college's file, a tenancy violation log is created immediately. This guarantees complete honesty and non-repudiation."*
JURY EVALUATION TAKEAWAY:	Searchable, tamper-proof audit log records every event with digital file fingerprints and Correlation IDs.

STEP 9: Final Approved Application State & Governance Impact • 9:45 – 10:30 (45s)	
ON-SCREEN ACTION & NAVIGATION CUES:	Display Delhi Tech application APP-202425-1-500123498-8769 showing solid green 'Approved' badge, active 6-step progress stepper, and 4 audit trail logs. Deliver confident concluding statement.
PRESENTER PITCH (FEATURE-FOCUSED & NATURAL LANGUAGE):	*"To conclude: look at Delhi Tech's completed application on screen. It has traveled the full statutory journey: from college submission through side-by-side OCR review, 4-pillar AI fraud detection, scrutiny verification, committee scheduling, and final Council Approval. You can see the solid green 'Approved' badge, the 6-step progress stepper, and the complete audit history below. SAGE transforms the 200-page AICTE handbook into an intuitive digital platform. By focusing on practical features — side-by-side inspection, automated checks, live compliance countdowns, and conflict-free committee scheduling — SAGE turns a multi-month paper headache into a fast, fair, and completely transparent digital experience. Thank you, and we welcome your questions."*
JURY EVALUATION TAKEAWAY:	Concludes on the final approved screen, proving complete end-to-end lifecycle governance.

5. Jury Q&A Defense Cheat Sheet (Clear & Practical Answers)

Straightforward, feature-grounded answers to expected jury and technical panel questions:

Q1: What happens if an applicant uploads a fake document or an AI-generated affidavit?

Clear Answer: *"SAGE's Dual-Pane Inspector automatically runs every uploaded document through our 4-Pillar Document Check. If an applicant uses ChatGPT to draft an affidavit or uploads a certificate without official stamp paper, the system flags the document in bold red with an alert badge and explains exactly what is wrong. The scrutiny officer can see this immediately and return the file with one click."*

Q2: Does SAGE approve applications automatically without human oversight?

Clear Answer: *"No, absolutely not. SAGE is designed around human accountability. The software performs OCR extraction, highlights dates, checks fonts, and identifies potential tampering, but the final determination requires an authorized Scrutiny Officer to review the file, enter official notes, and click 'Mark Verified Clean' or 'Flag as Deficient'."*

Q3: How does the system make sure inspection committee members are completely impartial?

Clear Answer: *"Our committee allocation engine enforces strict statutory rules: it checks each professor's registered home state and alma mater, and automatically excludes them from inspecting any institution in that state. It also guarantees exactly three qualified professors per panel and strictly excludes Sundays from campus visits."*

Q4: Can an institution replace a bad document and erase their mistake?

Clear Answer: *"No. SAGE uses immutable document version control. When an institution uploads a corrected Version 2 certificate, the original Version 1 file remains permanently archived in the audit drawer with its upload timestamp, the officer who uploaded it, and its digital file fingerprint. The full evidence history remains visible to AICTE officers."*

Q5: Can one college view or tamper with another college's application data?

Clear Answer: *"No. All data access is strictly isolated at the database level by institutional identity. If a college user tries to view or alter records belonging to another institution, the system blocks the request immediately with an access denied error and creates a security violation log in the audit trail."*